

Self-Similarity in Online Networks During Social Movements

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Online platforms provide an infrastructure for social movements, leaving digital traces that can be modelled as networks to quantify how information, participation, and coordination emerge during episodes of collective action and evolve over time. In this work, we unveil the emergence of scale-invariant **connectivity structure during** social movements through the analysis of three geographically and sociopolitically distinct massive mobilization events. By constructing co-occurrence networks from Twitter (now X) hashtag data and applying a degree-thresholding renormalisation procedure, we demonstrate that these highly correlated social phenomena exhibit clear signatures of self-similarity at peak mobilisation times. These critical points are characterised by modular-to-nested transitions, both in the co-occurrence networks and the bi-partite ones, maxima in user participation, and clustering spectrum collapse across multiple network scales. Furthermore, the results hint at the emergence of a latent metric structure that supports successful hyperbolic embedding, providing an estimate of effective **distances that support efficient communication**. Together, these findings suggest that self-similarity may constitute a universal organising principle of social movements during peak mobilisation phases.

I. INTRODUCTION

Cooperative social phenomena in online networks are short-lived bursts of coordinated behaviour in which many users align around a shared target or theme. This produces high correlations in activity. Users post and interact in synchronised bursts, reuse the same language, hashtags, memes, and URLs, and engage with the same accounts. This creates overlapping interaction patterns due to the shared goal of attracting attention and achieving a common objective within the short time span of an event. Large-scale studies have connected online social media behaviors to offline protest, showing that increased coordination of messages on social media such as Twitter using specific hashtags is associated with subsequent large-scale decentralised protests [1–3]. In network terms, structural signatures of these episodes include temporally clustered edges, denser and more cohesive communities, greater centralisation around hubs, and more similar user interaction profiles, with the coordination signal fading once the event window closes [4–6].

Intuitively, the structure of online social networks during peak mobilisation phases may suggest the emergence

of critical-like states in which collective action becomes highly correlated across users and time [6], so that small triggers can cascade into system-wide bursts of activity. However, these critical-like signatures remain largely uncharacterised. In particular, scale invariance and other statistical scaling properties in multiscale network organisation during these mobilisation episodes remain largely unexplored at the topological level, and the relatively scarce existing work has mostly focused on dynamical characterisation and modelling. [7, 8].

In complex networks, structural scale invariance has been investigated as self-similarity across length scales, adapting Mandelbrot’s ideas to characterise certain geometric objects known as fractals [9, 10], which express the invariance of form under length rescaling. Such rescaling has been implemented in networks using a variety of definitions of internode distance [11]. For instance, shortest path length was originally used in combination with box-counting coarse-graining to explore fractality and define fractal dimensions in networks [12, 13]. However, **using shortest-path length as a distance measure has serious limitations**. Because complex networks contain long-range connections, shortest-path lengths are narrowly distributed around a well-defined mean that is typically very small relative to the number of nodes. For instance, an average of 3.57 connections has been

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reported among millions of Facebook users [14]. Consequently, the small-world property restricts the range of available path-length scales, hindering the definition of concepts such as locality and distinguishability. Hence, other approaches have been introduced.

Alternatively, Serrano et al. introduced a family of network models in hidden metric space, known as the $\mathbb{S}^D$ model, in which the probability that two nodes are connected decreases with their effective distance [15]. This latent geometry was subsequently shown to be effectively hyperbolic through the isomorphism between the $\mathbb{S}^D$ and $\mathbb{H}^{D+1}$ models [16]. Since then, a variety of network-embedding methods have been developed to produce maps of real networks in hyperbolic space [17–21]. This geometric approach has revealed self-similarity across scales as a fundamental symmetry of real networks, implying that the same connectivity law governs their structure at different scales. This property is explained by the renormalizability of the $\mathbb{S}^D$ model, expressed through the statistical equivalence of subgraphs generated by renormalization transformations.

One such transformation is geometric renormalisation, which has shown that real networks exhibit self-similar structure across length scales [22–25]. Another is a simple degree-thresholding renormalisation (DTR) scheme, which generates a nested hierarchy of subgraphs by progressively removing nodes whose degrees fall below a prescribed threshold. Many real-world networks display self-similarity along the DTR flow, once the nodes’ degrees in the subgraphs are rescaled by the average degree of the corresponding subgraph. This symmetry, referred to here as multiscale k -cut self-similarity, together with high levels of clustering, provides a strong signature of an underlying spatial organization. As proven in [26], these features are inherent to geometric random graphs embedded in Riemannian manifolds of any constant curvature and in any dimension, provided that the nodes have a scale-free degree distribution and interactions are pairwise.

Motivated by the fact that online social networks are already characterised by high levels of clustering due to the triadic closure mechanism that drives human interactions [27], in this work we explored the **k-cut self-similarity** implied by latent geometry in complex networks. More precisely, we used the **DTR transformation** to search for signatures of multiscale k -cut self-similarity in the structural features of online social networks in the context of social movements. We found clear evidence of self-similar structural patterns emerging at peak mobilisation times across diverse geographical and sociopolitical contexts, suggesting that multiscale k -cut self-similarity may act as an ordering principle during highly correlated and clustered social events. We therefore adopt the geometric modeling framework and use hyperbolic embeddings to obtain a geometric representation of the hashtag co-occurrence networks, thereby defining effective distances between hashtags. We further show that greedy routing on these embeddings achieves high

navigability during critical mobilization times indicating enhanced communicability as compared with non-critical time windows. **TO BE DONE.**

The remainder of this paper is organised as follows. We continue with section II, where we present the events for which the data were collected, as well as the methodology used to gather those data. Then, in Section III, we report the empirical results and provide a comprehensive analysis of the observed outcomes obtained through the application of the methodology described in the section *Methods*. Finally, in Section IV, we summarise the main findings and implications of our work.

II. DATASETS

We analysed data collected from Twitter (now X) covering three geographically and sociopolitically distinct episodes of large-scale mobilisation. In each case, online activity surged in the days surrounding the event and culminated in collective street action. Two datasets correspond to Argentina. The first, *No al Tarifazo*, relates to a wave of nationwide protests in early January 2019 against sharp increases in public utility and transport tariffs announced by the government at the end of 2018; the demonstrations were organised through social media and took the form of simultaneous *cacerolazos* (pot-banging protests) across the country [28]. The second, *9n*, refers to the mobilisation of 9 November 2019, in which citizens demanded that the independence of the judiciary be preserved amid the impending change of national government. The two Argentinian events differ markedly in nature: *No al Tarifazo* was a socio-economic protest, driven by grassroots neighbourhood organisations and centred on the cost of essential services, whereas *9n* concerned an institutional matter: the independence of the judiciary, and mobilised different sectors of society [29]. They thus represent contrasting types of collective action rather than two instances of the same phenomenon. The third dataset, *Charlie Hebdo*, relates to France in the aftermath of the terrorist attack on the offices of the satirical magazine *Charlie Hebdo* in Paris on 7 January 2015, which triggered a mass demonstration in defence of freedom of expression on 11 January 2015 [30].

For each social movement, we identified the two most frequently used hashtags during the demonstration day(s) and compiled the set of users who posted at least one tweet containing either hashtag during that period. We then collected all tweets posted by these users over a broader observation window. Finally, we extracted from those tweets every hashtag appearing in their messages. Tweets for the two Argentine mobilisations were restricted to those originating from Argentina, whereas tweets for the Charlie Hebdo dataset were restricted to those originating from France. Table I summarises the seed hashtags used for data collection, and basic dataset statistics (number of users and number of distinct hash-

TABLE I: Description of the datasets employed.

Dataset	Timespan of Data Collection	Seed hashtags	# hashtags	# users	Country
<i>No al tarifazo</i>	01/01/2019 6h – 07/01/2019, 23h	#noaltarifazo and #ruidazonacional	136433	9670	Argentina
<i>9n</i>	08/11/2019, 6h – 10/11/2019, 20h	#9n and #9ngranmarchaporlajusticia	219488	8022	Argentina
<i>Charlie Hebdo</i>	10/01/2015, 1h – 12/01/2015, 23h	#charliehebdo and #jesuischarlie	185670	16146	France

tags).

III. RESULTS

A. Degree-thresholding renormalisation procedure

Once all tweets were collected, we aggregated their associated hashtags into Temporal Windows (TWs), which consist on all the hashtags posted between their start and end times. The TW duration for each dataset was selected to maximise the resolution of the modular-to-nested transition, which defines the critical regime in [6], and has also been characterised through entropic measures in [31]. The modular-to-nested transition is manifested as a sharp decrease in modularity accompanied by an increase in nestedness; see *Methods* for more details and the explanation of how these metrics were measured. We tested aggregation intervals ranging from one to six hours. Based on this exploration, tweets related to the *9n* mobilisation were aggregated into one-hour windows, whereas those associated with the No al Tarifazo and Charlie Hebdo movements were aggregated into two-hour windows.

Once all tweets were collected, we aggregated their associated hashtags into Temporal Windows (TWs), each consisting of all the hashtags posted between its start and end times. For every dataset we first identified the critical hour, at which the modular-to-nested transition occurs. This transition defines the critical regime [6] and has also been characterised through entropic measures [31]. The critical temporal window is the TW containing that hour. Its width was fixed by a single criterion: the shortest width whose hashtag co-occurrence network reaches at least 700 unique hashtags. This threshold guarantees sufficient statistical mass for the renormalisation analysis (see *Methods*). Because the datasets differ in density, this criterion yields a one-hour window for the *9n* mobilisation and two-hour windows for the No al Tarifazo and Charlie Hebdo movements.

For each TW, we constructed weighted bipartite networks linking hashtags and users, where the weight of each edge represents the number of times a given user employed the corresponding hashtag during that temporal interval. We also built hashtag co-occurrence networks where hashtags are represented as nodes, and an undirected edge was added between any pair of hashtags co-used by at least one user within the same TW. Edge weights in the co-occurrence network were defined as the number of distinct users who posted both hash-

tags during that period, thus providing a user-based measure of co-occurrence strength. *I would move the paragraphs above to the previous section Datasets and rename it something like "From data to networks". I would not use a subsection Results.*

B. Detecting the Critical Temporal Window

Figure 2a-c shows the number of unique hashtags and unique users across TWs for the three events. Because these networks reflect online activity within each window, we highlight in gray the intervals between 01:00 and 08:00 local time in the respective countries where each social movement took place. These low-activity TWs contain sparse participation and limited relevant content, and are therefore excluded from further analysis. Within the active periods, there is a single clear dominant peak of engagement in *No al tarifazo*, while *Charlie Hebdo* and *9n* display several maxima associated with moments of maximum online activity.

For each mobilisation, we computed the evolution of modularity and nestedness across all TWs during the period of analysis. The results are reported in Fig. 2d-f for the bipartite user-hashtag networks and in Fig. SI.1 for the unipartite hashtag co-occurrence network. While panels (a-c) recompute activity statistics, and panels (d-f) recompute the modularity-nestedness analysis, also reported in Ref. [31] for the same three events, the analysis here uses the bipartite user-hashtag network rather than the unipartite hashtag co-occurrence network used in that study.

We next explored the multiscale organization of the hashtag co-occurrence networks in each TW, applying the DTR procedure introduced in [15]. This method compares the connectivity structures of the original and a subgraph network at a specific scale, once the degrees of nodes in the subgraph are rescaled by its average degree. More specifically, given a graph G , DTR produces a nested hierarchy of n subgraphs $G(k_T) \subseteq G$ obtained by removing all nodes from G whose degree is equal to or less than k_T , where $k_T = 0, 1, 2, \dots, n$ denotes the sequence of degree thresholds effectively filtering out less connected nodes. Figure 1 illustrates this procedure with a toy example. To quantify the similarity between the original and decimated networks, we used the ϵ^2 collapse-quality metric introduced in [26] (see *Methods* for details). The ϵ^2 is computed by averaging the squared relative differences between the degree-dependent statistical curves of the original network and those of the corresponding

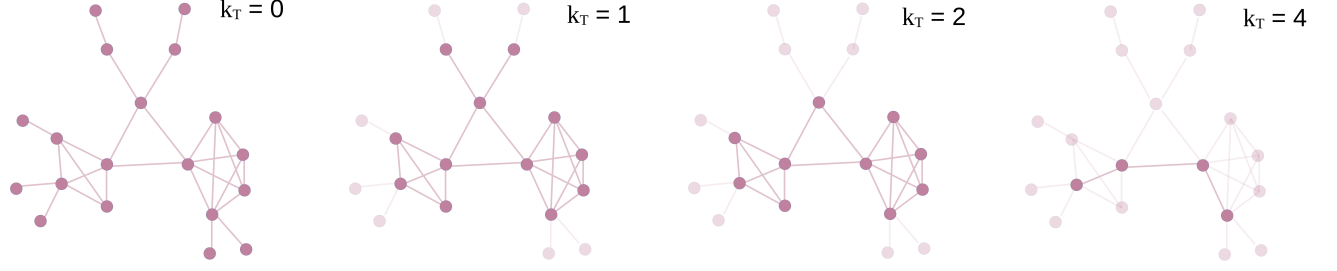


FIG. 1: **Degree-thresholding renormalisation procedure.** We can see the original network ($k_0 = 0$) and three subsequent renormalised ones $k_1 = 1$, $k_2 = 2$ and $k_3 = 4$, where the nodes with k_T links or less were removed from the network at the T -th iteration.

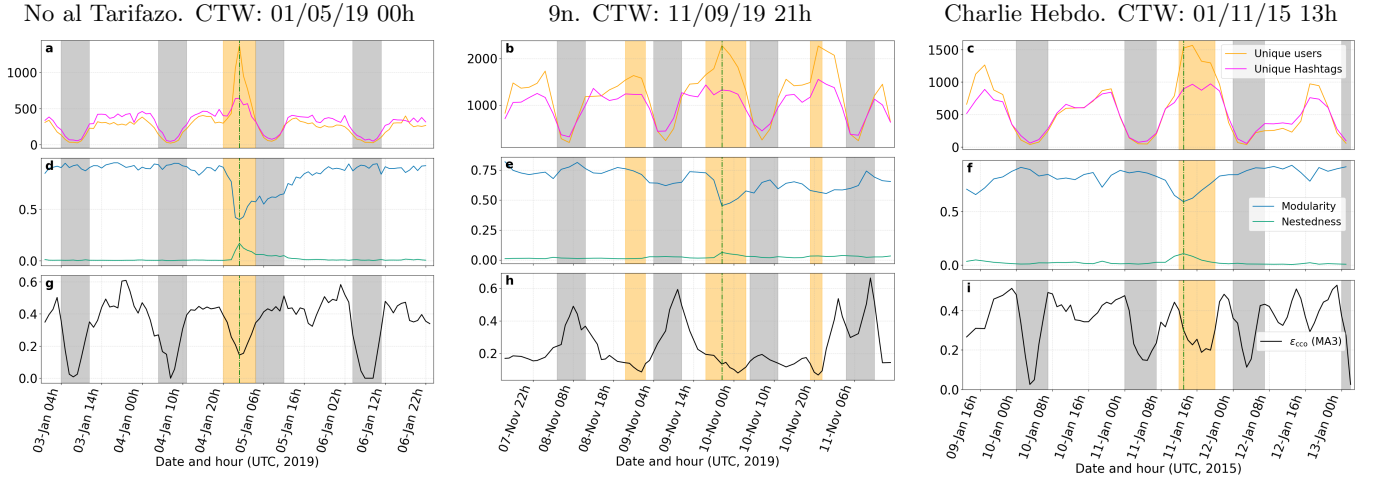


FIG. 2: **Detecting the Critical Temporal Window (CTW).** (a-c) Number of unique users and unique hashtags across temporal windows for the three social movements. Grey bands indicate low-activity periods (01:00–08:00 local time) excluded from analysis. (d-f) Evolution of modularity (blue) and nestedness (green) in bipartite user–hashtag networks. (g-i) Temporal evolution of the $\epsilon_c^2(k)$ self-similarity metric, computed as a moving average over $N = 3$ consecutive windows. Yellow bands in all panels highlight intervals with minimum $\epsilon_c^2(k)$ values that coincide with engagement peaks and modular-to-nested transitions, defining the Critical Temporal Windows. The identified CTW for each movement is marked by a green segmented line. For *No al Tarifazo* (left) and *Charlie Hebdo* (right), a single clear CTW emerges. For *9n* (centre), three candidate segments show low $\epsilon_c^2(k)$ values, but only the second exhibits a clear transition in the bipartite networks, unambiguously identifying the CTW. All times are given in UTC.

subgraphs over the range of rescaled degrees shared by both. *OJO notation epsilon! Which curves exactly did you used? Just clustering?*

Prior to the appliance of DTR, we filtered each network to remove hashtags tweeted by fewer than $N = 3$ users for *No al tarifazo* and *Charlie Hebdo* and $N = 5$ users for *9n*, thereby reducing noise while preserving the core structural features. Table SI.1 presents the percentage of nodes and edges retained after this filtering process for each dataset. *Check that this text is well placed here.* Fig. 2g-i shows the temporal evolution of $\epsilon_c^2(k)$ for each movement, computed as a moving average over $N = 3$ consecutive TWs to reduce noise. The raw ϵ^2 computed without smoothing are provided in Fig. SI.3 for all

three datasets. The unfiltered series exhibit considerable temporal fluctuations, with alternating local minima and maxima that obscure the global structure. This motivates the application of the three-window moving average: by attenuating high-frequency noise, the smoothing procedure enables the robust identification of global minima, without introducing spurious features or obscuring genuine critical windows.

The minima in $\epsilon_c^2(k)$ identify candidate CTWs, that is, TWs with maximal multiscale self-similarity. For *No al Tarifazo* and *Charlie Hebdo*, a single clear minimum emerges, which coincides precisely with engagement peaks (Fig. 2g-i) and well-defined modular-to-nested transitions in the bipartite networks (Fig. 2d-f), unam-

biguously defining unique candidate CTWs highlighted in yellow across all panels.

For *9n*, however, three segments show low $\epsilon_c^2(k)$ values (Fig. 2h). Notice that unipartite co-occurrence networks (Fig. SI.1) do not help to resolve this ambiguity, as they also exhibit two candidate windows with modular-to-nested transitions, making it difficult to unambiguously identify the CTW. The bipartite networks (Fig. 2h), however, resolve this ambiguity: only the second segment exhibits both a sharp drop in $\epsilon_c^2(k)$ and a clear modular-to-nested transition, unambiguously identifying Saturday, November 9, 2019 from 18:00 to 20:00 (Buenos Aires time, GMT−03:00) as the CTW (marked with a dashed line in all panels of Fig. 2).

For the *No al tarifazo* movement, the transition spans two consecutive TWs in the co-occurrence representation; however, the bipartite networks resolve it into a single, well-defined peak, allowing us to identify the CTW as Friday, January 4, 2019 from 21:00 to 22:00 (Buenos Aires time, GMT−03:00). Finally, using the same reasoning, for the *Charlie Hebdo* dataset, we set the CTW to Sunday, January 11, 2015 from 14:00 to 16:00 (Paris time, GMT+01:00). The bipartite representation thus provides a clearer signal for detecting the CTWs and is used here as the primary guide for discerning the transitions.

We calculated the main topological properties of each subgraph in the DTR flow of the networks in the identified CTWs. In Fig. 3a-c, we present the complementary cumulative degree distribution function (CCDF), which quantifies the probability that a node has degree no smaller than a certain value, for a selection of DTR subgraphs of each dataset in its CTW. For each curve, the x-axis shows the degree rescaled by the average degree of the corresponding subgraph, which we call k_{res} . The insets show how the average degree of the resulting subgraphs increases with the applied threshold k_T until it reaches a maximum (marked by a vertical dashed line), after which it decreases due to finite-size effects. We therefore stop the sequence of subgraphs at this maximum. Although the curves span a limited range of values, their collapse is clearly evident.

We also calculated the clustering spectrum $\bar{c}(k)$ by averaging the local clustering coefficients $c_i = 2T_i/(k(k-1))$ of nodes in the same degree class k , where T_i is the number of triangles in which a node i participates. In Fig. 3d-f, we show the clustering spectrum of the subgraphs as a function of the degree rescaled by the average degree of the corresponding subgraph. Remarkably, we observe a clear collapse of the clustering coefficient onto a single functional form, across renormalisation steps for all three social movements. This collapse demonstrates the presence of a robust self-similar behaviour in the network structure over multiple scales, where local patterns of organisation are replicated at progressively larger scales.

The slight variations in the range of k_T values reflect differences in network size and density among the three events, yet the fundamental self-similar behaviour

remains consistent. Note that the original curve deviates slightly, indicating distinct statistical behaviour for the lowest-degree nodes.

The observed approximate scale-invariant behaviour suggests that the underlying organisational principles governing hashtag co-occurrence during critical mobilisation periods are largely independent of degree scale and persist across different levels of degree thresholding. Crucially, this property is specific to the CTW: as shown in Fig. SI.3, the CCDF and clustering spectrum curves fail to collapse for non-critical temporal windows across all three datasets, with ϵ^2 values substantially higher than those observed at the CTW, strongly suggesting that self-similarity is a distinctive structural signature of peak mobilisation.

C. Network embedding in hyperbolic space

The co-occurrence of clustering and the multiscale k -cut self-similarity in the network reconstructions of each dataset at the CTWs points to connectivity linked to geometric constraints as a potential explanation for the observed symmetry [26]. Here we use the geometric $\mathbb{S}^1$ network model, which generates networks with clustering and multiscale k -cut self-similarity while nodes have a popularity coordinate directly controlling observed degrees and a similarity coordinate implemented as an angular position in a one-dimensional sphere or circle. Connections are formed with likelihoods that increase with popularity and decrease with angular distance. Equivalently, the model has a pure geometric version in the hyperbolic plane, the $\mathbb{H}^2$ model, where the probability of connection solely depends on hyperbolic distances [16].

We employed the model-based D-Mercator tool [21] to embed the CTW networks into the hyperbolic plane. Taking the network topology as input, D-Mercator calculates the popularity and similarity coordinates of the nodes that give the maximum congruence between the $\mathbb{S}^1/\mathbb{H}^2$ model and the observed topology.

Figure 4 displays the hyperbolic embeddings of the CTW networks for the three social movements alongside the comparison of their empirical and inferred corresponding connection probability curves. The upper row shows the two-dimensional hyperbolic embeddings, where node size represents degree and spatial proximity in the hyperbolic plane reflects the likelihood of connection. In each embedding, the seed hashtags—such as *#noaltarifazo* and *#ruidazonacional*, *#9ngranmarchaporlajusticia*, and *#jesuischarlie* and *#charliehebdo*—occupy central positions with the lowest radial coordinates, reflecting their high popularity and role as focal points of collective attention. Surrounding them, thematically related hashtags (e.g. *#sepuede*, *#marcherepublicaine*, *#libertéégalitéfraternité*) cluster at small angular distances, indicating high similarity, whilst more peripheral or loosely related hashtags are pushed towards the outer regions of the hyperbolic

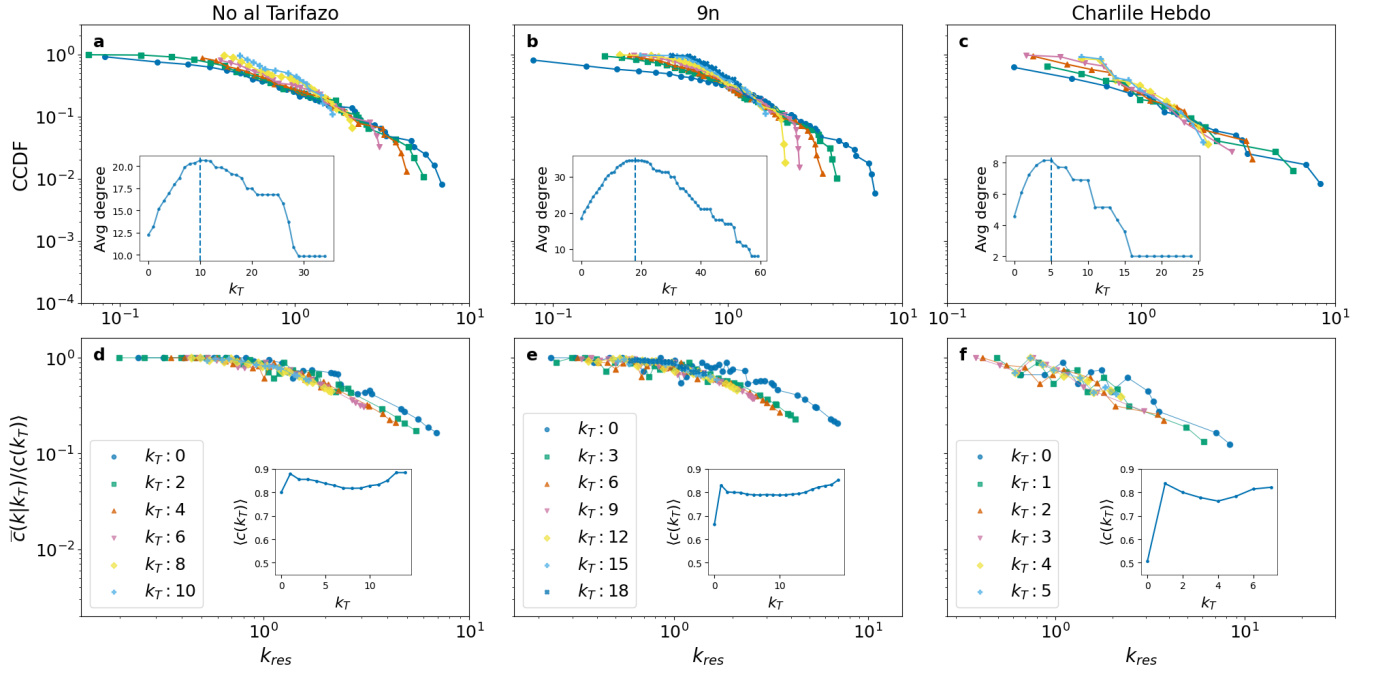


FIG. 3: **Complementary cumulative degree distribution and average clustering coefficient in the CTW.** The top row presents the complementary cumulative degree distribution function (CCDF) as a function of the rescaled degree, k_{res} , for various values of the degree threshold k_T . The inset illustrates how the average degree of the resulting networks increases with k_T . A blue dashed line indicates the cut-off point beyond which the results are excluded from further analysis due to finite-size effects forcing the average degree to decrease. The bottom row displays the spectrum of the average clustering coefficient rescaled by the average clustering coefficient of the corresponding subgraph as a function of the rescaled degree.

disc. The lower row presents the connection probability as a function of rescaled hyperbolic distance, comparing the inferred model (dashed line) against the empirical network data (black points). For all three datasets, we observe excellent agreement between the inferred hyperbolic geometry and the original network structure. The connection probability decays smoothly with hyperbolic distance, closely following the expected behavior for networks embedded in underlying metric spaces. This successful embedding provides strong evidence that our co-occurrence networks can be explained by a latent geometric structure satisfying the triangle inequality. Moreover, the successful embedding across all three geographically and sociopolitically diverse movements further substantiates the universality of self-similar organisation in social mobilisation. Regardless of the specific context—whether protests against government policies in Argentina or solidarity marches following a terrorist attack in France—the underlying network structures exhibit consistent geometric properties that facilitate efficient information dissemination and rapid coordination among participants.

IV. DISCUSSION

In this work, we have investigated the emergence of self-similar structural patterns in social movements through the analysis of three distinct mobilisation events: the 9n justice reform protests and No al Tarifazo demonstrations in Argentina, and the Charlie Hebdo marches in France. By applying the DTR degree-thresholding renormalisation procedure to co-occurrence networks constructed from [Twitter \(now X\)](#) data, we have demonstrated that these highly correlated social phenomena exhibit clear signatures of self-similarity at critical periods. The observation that self-similarity emerges across all three datasets, despite their differing geographical and socio-political contexts, points towards a universal characteristic of social movements during critical periods.

Our methodology successfully identified Critical Temporal Windows (CTWs) for each movement. The convergence of multiple indicators—including the maxima in user and hashtag participation, the modular-to-nested transition (in both bipartite and co-occurrence networks), and the ϵ^2 test for features collapse—provides robust evidence that these CTWs represent distinct phases in the dynamics of social mobilisation. Notably, the identified CTWs align with the moments when digital activity translated into massive physical demonstrations, suggest-

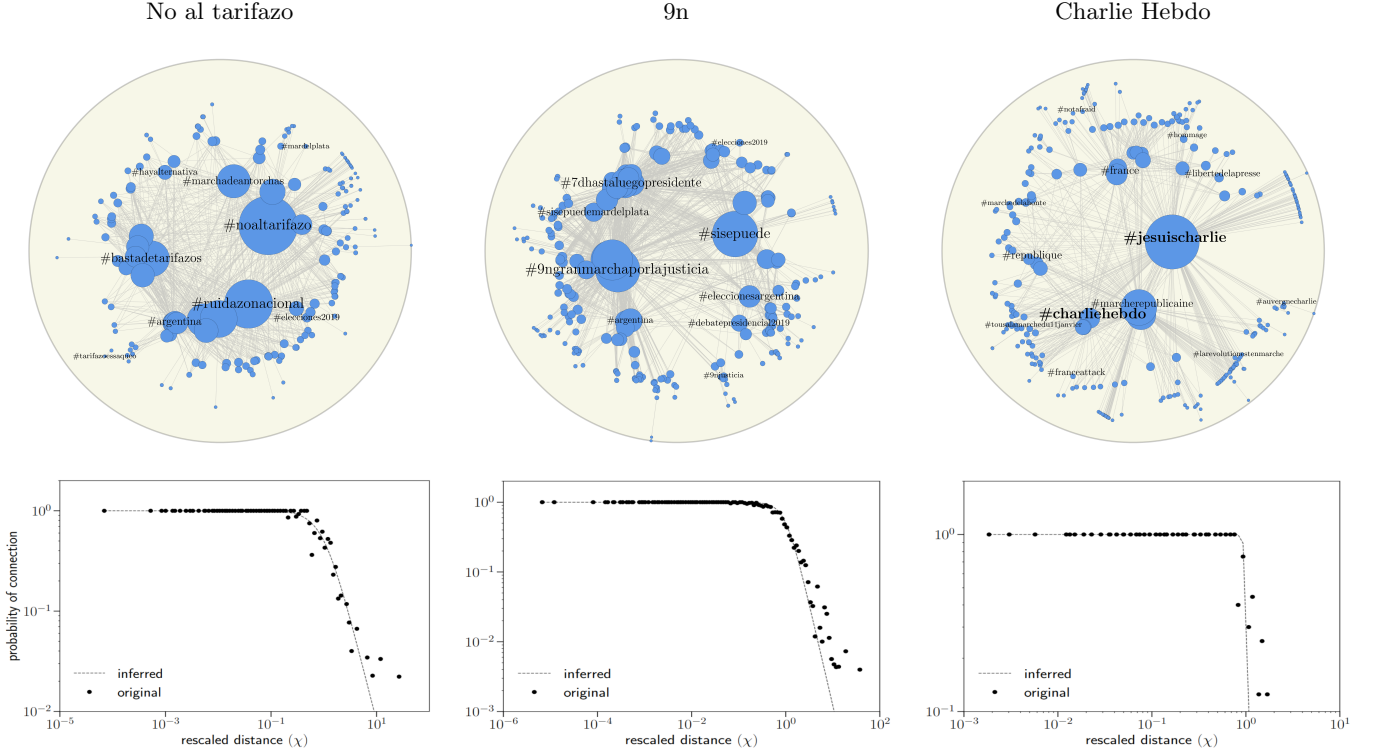


FIG. 4: Hyperbolic embeddings of the hashtag co-occurrence networks and connection probability profiles. The top row presents the two-dimensional projections of the hyperbolic embeddings obtained using the D-Mercator method for the filtered Critical Temporal Window networks of the three social movements. Node size is proportional to degree, and spatial proximity reflects the likelihood of connection in the underlying hyperbolic geometry. The bottom row displays the connection probability as a function of rescaled hyperbolic distance for each dataset. Black points represent the inferred model, whilst grey points correspond to the original network data. The excellent agreement between the model and empirical data demonstrates that the co-occurrence networks possess latent metric structures consistent with hyperbolic geometry, where connection probability decays smoothly with hyperbolic distance. From left to right: No al Tarifazo, 9n, and Charlie Hebdo datasets. Prominent hashtags such as #noaltarifazo, #9ngrammarchaporlajusticia, and #jesuischarlie are visible near the centre of each embedding, consistent with their high degree and central role in the mobilisation.

ing that the underlying network structures reflect fundamental organisational principles of collective attention.

In this context, "organisational principles of collective attention" refers to the self-organisation of attention within the network structure, that is, the modular-to-nested reorganisation and the self-similar connectivity that emerge when attention concentrates. It does not refer to the logistical organisation of the demonstrations. We make no causal or predictive claim regarding the physical events. The alignment we report is a temporal co-occurrence between the window of peak online attention and the mobilisation, and not an assertion that the network signature drives or precedes it.

The highly clustered nature of social interactions, driven by triadic closure mechanisms, is reflected in high levels of clustering. The clustering coefficient curves, as well as the degree distributions, collapse onto single curves obtained for the subgraphs generated by the DTR renormalisation procedure. Our interpretation is

that this scale-invariant behaviour emerges from the coordinated efforts of participants to achieve common goals within compressed time frames. The simultaneous occurrence of high levels of clustering and DTR self-similarity at the CTWs reduces the portfolio of models that can be used to explain these observations. On the other hand, one would like the models to be maximum entropy models, such that networks in the corresponding ensemble are maximally random except for the assumed constraints. As far as we know, the $\mathbb{S}^D/\mathbb{H}^{D+1}$ is the only family of models that produce networks with high levels of clustering and DTR self-similarity while being maximally random ensembles, as proved in [26] and [32].

Hence, our findings suggest that these networks possess a latent hyperbolic metric structure at critical periods. The successful embeddings using the D-Mercator method reinforces the notion that in these systems the likelihood of interaction decreases with an effective distance, which combines similarity and popularity dimensions. On these

hyperbolic maps, greedy routing exhibits high navigability during periods of critical mobilization, indicating enhanced communicability compared with non-critical time windows. **TO BE DONE.**

These results have important implications for understanding the organisational dynamics of social movements. The presence of self-similarity indicates that local patterns of coordination and information diffusion are replicated at progressively larger scales. This hierarchical self-organisation may represent an optimal structural configuration for balancing information dissemination with maintaining network coherence during critical periods. Our results suggest that analysing other social phenomena characterised by peak correlation in attention and collective actions would be valuable for advancing our understanding of the fundamental principles governing criticality in social systems.

METHODS

A. Data Gathering

Twitter/X data for the Argentine events (No al Tarifazo and 9N) were collected by a collaborator through the Twitter/X Academic Research API, as part of a separate published study [6]. Data for the Charlie Hebdo event were collected independently, via the same academic API but at a different time, and were subsequently acquired commercially for this study.

B. Modular-to-nested transitions

We characterise the modular-to-nested transition through two complementary quantities: the modularity of the network's community partition and its nestedness. Modularity measures how well a partition divides the network into internally dense, mutually sparse communities. For a graph $G = (V, E)$, with V the set of nodes and E the set of edges, it is defined as [33]

$$Q = \frac{1}{2m} \sum_{ij} \left(A_{ij} - \frac{k_i k_j}{2m} \right) \delta_{g_i, g_j}, \quad (1)$$

where A_{ij} is the weight of the edge connecting nodes i and j , and $A_{ij} = 0$ otherwise; $k_i = \sum_j A_{ij}$ is the weighted degree of node i ; $m = \frac{1}{2} \sum_{ij} A_{ij}$ is the total edge weight; δ_{g_i, g_j} is the Kronecker delta, which takes value 1 if $g_i = g_j$ and 0 otherwise; and g_i is the community assigned to node i .

The partition is the one that maximises Q , found with the Louvain algorithm [34] as implemented in NETWORKX [35]. Louvain is a heuristic that optimises Q

through iterative local node moves; a fixed random seed is used to ensure reproducibility.

To quantify nestedness, we use the NODF (Nestedness based on Overlap and Decreasing Fill) [36, 37]. By doing so, we quantify the extent to which the interactions of a node form subsets of the interactions of the higher-degree ones. Specifically, in undirected networks NODF is calculated as follows:

$$\text{NODF}_{uni} = \frac{2}{n(n-1)} \sum_{\substack{i < j \\ k_i \neq k_j}}^n \frac{O_{ij}}{\min(k_i, k_j)} \quad (2)$$

The quantity $O_{i,j} = \sum_{k=1}^n A_{ik} A_{jk}$ measures the number of common neighbours between each pair of nodes i and j using the overlap matrix $O_{i,j}$.

In this way, we obtain the average, over all pairs of nodes, of the overlap normalised by the lesser degree of the pair. Then, normalising the overlap of neighbours by the smaller degree captures hierarchy, because it tests whether the neighbours of the less-connected node form a subset of the more-connected ones. For example, if all of the smaller node's neighbours are also linked to the larger one, the ratio equals 1 (perfect nestedness). While if only some are shared, it lies between 0 and 1. Then, the average of this measure across all node pairs quantifies how much the network approaches the ideal nested (hierarchical) structure, where low-degree nodes are entirely embedded into the neighbourhoods of high-degree ones. Prior to computing nestedness, the weighted bipartite and co-occurrence networks were binarised by setting all non-zero edge weights to one. For both the unipartite co-occurrence and bipartite user-hashtag networks, Eq. (2) was applied to the full $n \times n$ binary adjacency matrix, where $n = R + C$ in the bipartite case, with R users and C hashtags.

C. ϵ^2 collapse-quality metric

To evaluate the collapse between the original distribution and those obtained after the procedure of degree-thresholding renormalisation, we computed the distance between them by means of the ϵ^2 collapse-quality metric. It quantifies the average squared relative difference between the curves representing structural properties of the original and renormalized networks. We leverage the definition introduced in [26] to calculate the values of $\epsilon_{\hat{c}(k)}$ as the average of the squared relative differences between each curve in the DTR flow and the one for original network across the selected topological property, using exponential binning with nbins number of bins within the shared range of the x-axis for the two curves:

$$\epsilon^2 = \frac{1}{n+1} \cdot \frac{1}{n_{bins}} \sum_{k_T} \sum_{i=1}^{n_{bins}} \left(\frac{\text{bin value}_i^{org} - \text{bin value}_i^{k_T}}{\text{bin value}_i^{org}} \right)^2.$$

Here, we used $n_{\text{bins}} = 20$. Smaller ϵ^2 values indicate a good collapse of the rescaled curves, i.e. self-similar behaviour. *OJO notation epsilon*.

D. Geometric model and network embedding

To investigate further whether the observed self-similarity at CTWs corresponds to an underlying geometric structure, we employed D-Mercator [21], a model-based embedding method that maps real networks into the $(D + 1)$ -dimensional hyperbolic space based on the $\mathbb{S}^D$ model [15]. D-Mercator implements a maximum-likelihood inference procedure that assigns each node two hidden coordinates: a popularity coordinate corresponding to a radial position in the latent hyperbolic space that correlates with node degree, and similarity coordinates corresponding to angular positions in a D -dimensional sphere that determine community structure. For simplicity, we used the two-dimensional version, where every node is characterized by a popularity variable or hidden degree κ_i and an angular position θ_i in the one-dimensional sphere representing the similarity space. In $D = 1$, D-Mercator infers the popularity-similarity coordinates of the $\mathbb{S}^1$ model [15] or, equivalently, the $\mathbb{H}^2$ formulation [16] in the hyperbolic plane, and has been shown to accurately reproduce complex network topologies whilst remaining computationally tractable. The method assumes that the connection probability between nodes i and j follows:

$$p_{ij} = \frac{1}{1 + \left(\frac{R\Delta\theta_{ij}}{\mu\kappa_i\kappa_j} \right)^\beta}, \quad (3)$$

where R is the radius of the similarity circle, $\Delta\theta_{ij}$ is the angular distance between the nodes, μ controls the average degree, and β controls the level of clustering quantifying the coupling between the network topology and the underlying geometry.

Prior to embedding, we filtered each CTW network to remove hashtags tweeted by fewer than N users, as low-activity nodes introduce noise that degrades inference quality. The threshold N was selected based on network size and density: after testing $N \in \{3, 5, 7, 10\}$, we determined that $N = 3$ for *No al tarifazo* and *Charlie Hebdo* and $N = 5$ for *9n* provides the optimal trade-off between noise reduction and preservation of structural features (see Table SI.1) for filtering statistics).

The quality of each embedding was assessed by comparing the empirical connection probability $P(\chi)$ as a function of rescaled hyperbolic distance with the theoretical prediction from the inferred model. High-quality embeddings exhibit smooth decay of $P(\chi)$ with distance, indicating that the network structure is well-explained by the latent hyperbolic geometry.

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